

Модуль 4. Практика 1

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Назначим компьютерам ip адреса в одной локальной сети

```
PC1> ip 192.168.1.2/24
```

Checking for duplicate address...

PC1 : 192.168.1.2 255.255.255.0

```
PC2> ip 192.168.1.3/24
```

Checking for duplicate address...

PC2 : 192.168.1.3 255.255.255.0

```
PC1> ping 192.168.1.3
```

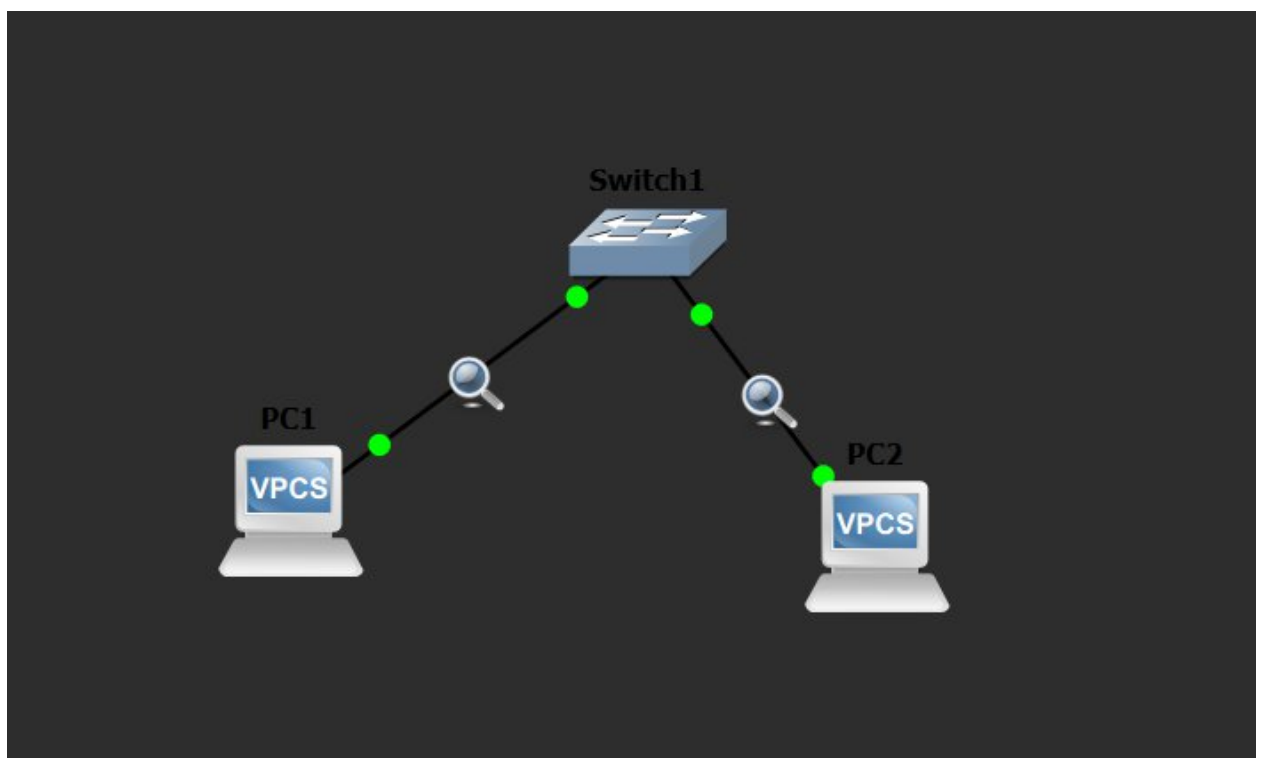
84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=0.232 ms

84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=0.243 ms

84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=0.284 ms

84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=0.234 ms

84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=0.242 ms



В Wireshark видим отправленные данные на соединении между PC1 и коммутатором. После ARP запросов видим ICMP запрос-ответы, это и есть наши пинги.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.1.3? Tell 192.168.1.2
2	0.000135	Private_66:68:01	Private_66:68:00	ARP	64	192.168.1.3 is at 00:50:79:66:68:01
3	0.001035	192.168.1.2	192.168.1.3	ICMP	98	Echo (ping) request id=0x25e9, seq=1/256, ttl=64 (reply in 4)
4	0.001139	192.168.1.3	192.168.1.2	ICMP	98	Echo (ping) reply id=0x25e9, seq=1/256, ttl=64 (request in 3)
5	1.002193	192.168.1.2	192.168.1.3	ICMP	98	Echo (ping) request id=0x26e9, seq=2/512, ttl=64 (reply in 6)
6	1.002318	192.168.1.3	192.168.1.2	ICMP	98	Echo (ping) reply id=0x26e9, seq=2/512, ttl=64 (request in 5)
7	2.003304	192.168.1.2	192.168.1.3	ICMP	98	Echo (ping) request id=0x27e9, seq=3/768, ttl=64 (reply in 8)
8	2.003429	192.168.1.3	192.168.1.2	ICMP	98	Echo (ping) reply id=0x27e9, seq=3/768, ttl=64 (request in 7)
9	3.004478	192.168.1.2	192.168.1.3	ICMP	98	Echo (ping) request id=0x28e9, seq=4/1024, ttl=64 (reply in 10)
10	3.004615	192.168.1.3	192.168.1.2	ICMP	98	Echo (ping) reply id=0x28e9, seq=4/1024, ttl=64 (request in 9)
11	4.005602	192.168.1.2	192.168.1.3	ICMP	98	Echo (ping) request id=0x29e9, seq=5/1280, ttl=64 (reply in 12)
12	4.005736	192.168.1.3	192.168.1.2	ICMP	98	Echo (ping) reply id=0x29e9, seq=5/1280, ttl=64 (request in 11)

PC1 отправляет широковещательные ARP по локальной сети с вопросом к комьютеру с IP 192.168.1.3 какой у него MAC адрес и указывает свой MAC адрес для ответа

```
▶ Frame 1: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
▶ Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
▼ Address Resolution Protocol (request)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: request (1)
  Sender MAC address: Private_66:68:00 (00:50:79:66:68:00)
  Sender IP address: 192.168.1.2
  Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
  Target IP address: 192.168.1.3
```

PC2 отправляет ответ PC1 со своим MAC адресом. Этот адрес записывается в таблицу MAC адресов у PC1 для дальнейшего использования

```
▶ Frame 2: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
▶ Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:66:68:00)
▼ Address Resolution Protocol (reply)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: reply (2)
  Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)
  Sender IP address: 192.168.1.3
  Target MAC address: Private_66:68:00 (00:50:79:66:68:00)
  Target IP address: 192.168.1.2
```

На соединении между PC2 и коммутатором тот же самый трафик

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.1.3? Tell 192.168.1.2
2	0.000134	Private_66:68:01	Private_66:68:00	ARP	64	192.168.1.3 is at 00:50:79:66:68:01
3	0.001031	192.168.1.2	192.168.1.3	ICMP	98	Echo (ping) request id=0x40ea, seq=1/256, ttl=64 (reply in 4)
4	0.001146	192.168.1.3	192.168.1.2	ICMP	98	Echo (ping) reply id=0x40ea, seq=1/256, ttl=64 (request in 3)
5	1.002212	192.168.1.2	192.168.1.3	ICMP	98	Echo (ping) request id=0x41ea, seq=2/512, ttl=64 (reply in 6)
6	1.002351	192.168.1.3	192.168.1.2	ICMP	98	Echo (ping) reply id=0x41ea, seq=2/512, ttl=64 (request in 5)
7	2.003369	192.168.1.2	192.168.1.3	ICMP	98	Echo (ping) request id=0x42ea, seq=3/768, ttl=64 (reply in 8)
8	2.003509	192.168.1.3	192.168.1.2	ICMP	98	Echo (ping) reply id=0x42ea, seq=3/768, ttl=64 (request in 7)
9	3.004499	192.168.1.2	192.168.1.3	ICMP	98	Echo (ping) request id=0x43ea, seq=4/1024, ttl=64 (reply in 10)
10	3.004676	192.168.1.3	192.168.1.2	ICMP	98	Echo (ping) reply id=0x43ea, seq=4/1024, ttl=64 (request in 9)
11	4.005639	192.168.1.2	192.168.1.3	ICMP	98	Echo (ping) request id=0x44ea, seq=5/1280, ttl=64 (reply in 12)
12	4.005790	192.168.1.3	192.168.1.2	ICMP	98	Echo (ping) reply id=0x44ea, seq=5/1280, ttl=64 (request in 11)

```

▶ Frame 1: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
▶ Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
▼ Address Resolution Protocol (request)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: request (1)
  Sender MAC address: Private_66:68:00 (00:50:79:66:68:00)
  Sender IP address: 192.168.1.2
  Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
  Target IP address: 192.168.1.3

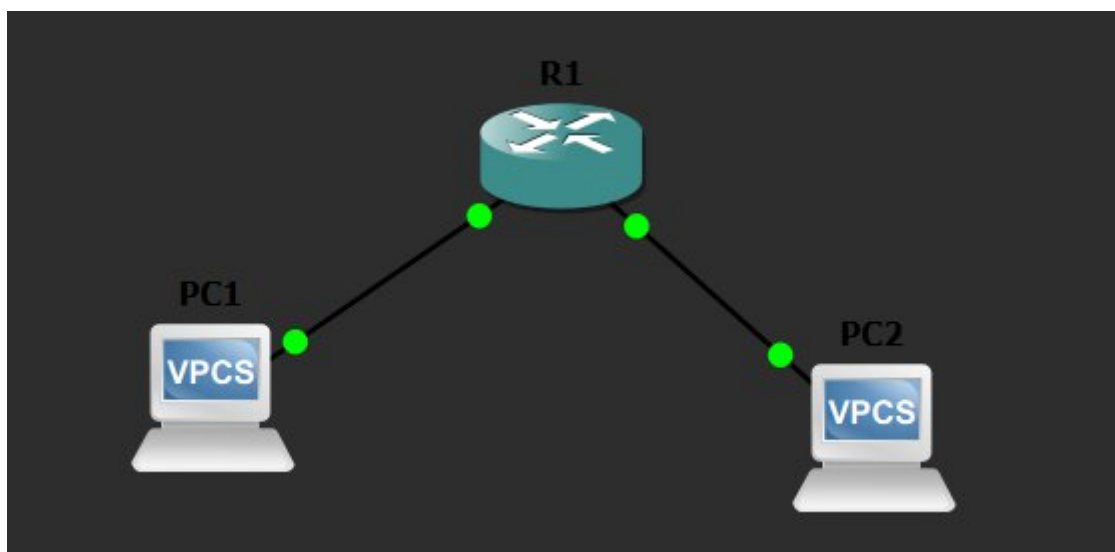
```

```

▶ Frame 2: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
▶ Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:66:68:00)
▼ Address Resolution Protocol (reply)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: reply (2)
  Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)
  Sender IP address: 192.168.1.3
  Target MAC address: Private_66:68:00 (00:50:79:66:68:00)
  Target IP address: 192.168.1.2

```

Выполним часть задания с маршрутизатором вместо коммутатора



Назначим интерфейсам роутера IP адреса из разных сетей

```
R1#configure terminal
```

```
R1(config)#interface ethernet 2/0
```

```
R1(config-if)#ip address 192.168.1.1 255.255.255.0
```

```
R1(config-if)#no shutdown
```

```
R1(config-if)#exit
```

```
R1(config)#exit
```

```
*Mar 1 00:06:04.827: %LINK-3-UPDOWN: Interface Ethernet2/0, changed state to up
```

```
*Mar 1 00:06:05.827: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/0, changed state to up
```

```
R1(config)#interface ethernet 2/1
```

```
R1(config-if)#ip address 192.168.2.1 255.255.255.0
```

```
R1(config-if)#no shutdown
```

```
R1(config-if)#exit
```

```
R1(config)#
```

```
*Mar 1 00:06:44.671: %LINK-3-UPDOWN: Interface Ethernet2/1, changed state to up
```

```
*Mar 1 00:06:45.671: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet2/1, changed state to up
```

```
R1#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	unset	administratively down	down
FastEthernet1/0	unassigned	YES	unset	administratively down	down
Ethernet2/0	192.168.1.1	YES	manual	up	up
Ethernet2/1	192.168.2.1	YES	manual	up	up
Ethernet2/2	unassigned	YES	unset	administratively down	down
Ethernet2/3	unassigned	YES	unset	administratively down	down
Serial3/0	unassigned	YES	unset	administratively down	down
Serial3/1	unassigned	YES	unset	administratively down	down
Serial3/2	unassigned	YES	unset	administratively down	down
Serial3/3	unassigned	YES	unset	administratively down	down

Назначим компьютерам адреса из этих сетей

```
PC2> ip 192.168.2.3/24 192.168.2.1
```

Checking for duplicate address...

PC2 : 192.168.2.3 255.255.255.0 gateway 192.168.2.1

PC1> ip 192.168.1.2/24 192.168.1.1

Checking for duplicate address...

PC1 : 192.168.1.2 255.255.255.0 gateway 192.168.1.1

PC1 сравнивает пингуемый адрес со своей сетью, понимает что нужный адрес находится в другой сети и делает ARP запрос IP адреса шлюза. После получения MAC адреса шлюза, PC1 отправляет запрос на этот адрес на канальном уровне, но на IP адрес PC2 на сетевом уровне. Маршрутизатор сравнивает IP адрес получателя со своей таблицей интерфейсов и сетей и определяет на какой именно интерфейс необходимо направить пакет. Но сначала он делает ARP запрос с IP адресом получателя и определяет MAC адрес PC2. Далее после того как PC2 получает ping он отправляет ответ по такому же алгоритму.

PC1-R1

48	303.727368	cc:01:0d:28:00:20	CDP/VTP/DTP/PagP/UDLD	CDP	343 Device ID: R1 Port ID: Ethernet2/0
49	304.193531	Private_66:68:00	Broadcast	ARP	64 Who has 192.168.1.1? Tell 192.168.1.2
50	304.200262	cc:01:0d:28:00:20	Private_66:68:00	ARP	60 192.168.1.1 is at cc:01:0d:28:00:20
51	304.200955	192.168.1.2	192.168.2.3	ICMP	98 Echo (ping) request id=0x63f1, seq=1/256, ttl=64 (reply in 52)
52	304.230478	192.168.2.3	192.168.1.2	ICMP	98 Echo (ping) reply id=0x63f1, seq=1/256, ttl=63 (request in 51)
53	305.230711	192.168.1.2	192.168.2.3	ICMP	98 Echo (ping) request id=0x64f1, seq=2/512, ttl=64 (reply in 54)
54	305.246445	192.168.2.3	192.168.1.2	ICMP	98 Echo (ping) reply id=0x64f1, seq=2/512, ttl=63 (request in 53)
55	306.246642	192.168.1.2	192.168.2.3	ICMP	98 Echo (ping) request id=0x65f1, seq=3/768, ttl=64 (reply in 56)
56	306.262514	192.168.2.3	192.168.1.2	ICMP	98 Echo (ping) reply id=0x65f1, seq=3/768, ttl=63 (request in 55)
57	307.263753	192.168.1.2	192.168.2.3	ICMP	98 Echo (ping) request id=0x66f1, seq=4/1024, ttl=64 (reply in 58)
58	307.278598	192.168.2.3	192.168.1.2	ICMP	98 Echo (ping) reply id=0x66f1, seq=4/1024, ttl=63 (request in 57)
59	308.279727	192.168.1.2	192.168.2.3	ICMP	98 Echo (ping) request id=0x67f1, seq=5/1280, ttl=64 (reply in 60)
60	308.294597	192.168.2.3	192.168.1.2	ICMP	98 Echo (ping) reply id=0x67f1, seq=5/1280, ttl=63 (request in 59)

```
Frame 4: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: cc:01:0d:28:00:20 (cc:01:0d:28:00:20)
Internet Protocol Version 4, Src: 192.168.1.2, Dst: 192.168.2.3
Internet Control Message Protocol
  Type: Echo (ping) request (8)
  Code: 0
  Checksum: 0xdd1a [correct]
  [Checksum Status: Good]
  Identifier (BE): 17136 (0x42f0)
  Identifier (LE): 61506 (0xf042)
  Sequence Number (BE): 1 (0x0001)
  Sequence Number (LE): 256 (0x0100)
  [Response frame: 5]
  Data (56 bytes)
```



```

▶ Frame 5: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
▶ Ethernet II, Src: cc:01:0d:28:00:20 (cc:01:0d:28:00:20), Dst: Private_66:68:00 (00:50:79:66:68:00)
▶ Internet Protocol Version 4, Src: 192.168.2.3, Dst: 192.168.1.2
▼ Internet Control Message Protocol
  Type: Echo (ping) reply (0)
  Code: 0
  Checksum: 0xe51a [correct]
  [Checksum Status: Good]
  Identifier (BE): 17136 (0x42f0)
  Identifier (LE): 61506 (0xf042)
  Sequence Number (BE): 1 (0x0001)
  Sequence Number (LE): 256 (0x0100)
  [Request frame: 4]
  [Response time: 16,891 ms]
▶ Data (56 bytes)

```

```

▶ Frame 49: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
▶ Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
▼ Address Resolution Protocol (request)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: request (1)
  Sender MAC address: Private_66:68:00 (00:50:79:66:68:00)
  Sender IP address: 192.168.1.2
  Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
  Target IP address: 192.168.1.1

```

```

▶ Frame 50: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
▶ Ethernet II, Src: cc:01:0d:28:00:20 (cc:01:0d:28:00:20), Dst: Private_66:68:00 (00:50:79:66:68:00)
▼ Address Resolution Protocol (reply)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: reply (2)
  Sender MAC address: cc:01:0d:28:00:20 (cc:01:0d:28:00:20)
  Sender IP address: 192.168.1.1
  Target MAC address: Private_66:68:00 (00:50:79:66:68:00)
  Target IP address: 192.168.1.2

```

R1-PC2

47	294.209211	192.168.1.2	192.168.2.3	ICMP	98 Echo (ping) request	id=0x63f1, seq=1/256, ttl=63 (reply in 50)
48	294.209363	Private_66:68:01	Broadcast	ARP	64 Who has 192.168.2.1? Tell 192.168.2.3	
49	294.219274	cc:01:0d:28:00:21	Private_66:68:01	ARP	60 192.168.2.1 is at cc:01:0d:28:00:21	
50	294.219914	192.168.2.3	192.168.1.2	ICMP	98 Echo (ping) reply	id=0x63f1, seq=1/256, ttl=64 (request in 47)
51	295.235290	192.168.1.2	192.168.2.3	ICMP	98 Echo (ping) request	id=0x64f1, seq=2/512, ttl=63 (reply in 52)
52	295.235391	192.168.2.3	192.168.1.2	ICMP	98 Echo (ping) reply	id=0x64f1, seq=2/512, ttl=64 (request in 51)
53	296.251341	192.168.1.2	192.168.2.3	ICMP	98 Echo (ping) request	id=0x65f1, seq=3/768, ttl=63 (reply in 54)
54	296.251432	192.168.2.3	192.168.1.2	ICMP	98 Echo (ping) reply	id=0x65f1, seq=3/768, ttl=64 (request in 53)
55	297.267391	192.168.1.2	192.168.2.3	ICMP	98 Echo (ping) request	id=0x66f1, seq=4/1024, ttl=63 (reply in 56)
56	297.267492	192.168.2.3	192.168.1.2	ICMP	98 Echo (ping) reply	id=0x66f1, seq=4/1024, ttl=64 (request in 55)
57	298.283394	192.168.1.2	192.168.2.3	ICMP	98 Echo (ping) request	id=0x67f1, seq=5/1280, ttl=63 (reply in 58)
58	298.283496	192.168.2.3	192.168.1.2	ICMP	98 Echo (ping) reply	id=0x67f1, seq=5/1280, ttl=64 (request in 57)

```

▶ Frame 48: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
▶ Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
▼ Address Resolution Protocol (request)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: request (1)
  Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)
  Sender IP address: 192.168.2.3
  Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
  Target IP address: 192.168.2.1

```

```

▶ Frame 49: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
▶ Ethernet II, Src: cc:01:0d:28:00:21 (cc:01:0d:28:00:21), Dst: Private_66:68:01 (00:50:79:66:68:01)
▼ Address Resolution Protocol (reply)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: reply (2)
  Sender MAC address: cc:01:0d:28:00:21 (cc:01:0d:28:00:21)
  Sender IP address: 192.168.2.1
  Target MAC address: Private_66:68:01 (00:50:79:66:68:01)
  Target IP address: 192.168.2.3

```

```

▶ Frame 50: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
▶ Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: cc:01:0d:28:00:21 (cc:01:0d:28:00:21)
▶ Internet Protocol Version 4, Src: 192.168.2.3, Dst: 192.168.1.2
▼ Internet Control Message Protocol
  Type: Echo (ping) reply (0)
  Code: 0
  Checksum: 0xc419 [correct]
  [Checksum Status: Good]
  Identifier (BE): 25585 (0x63f1)
  Identifier (LE): 61795 (0xf163)
  Sequence Number (BE): 1 (0x0001)
  Sequence Number (LE): 256 (0x0100)
  [Request frame: 47]
  [Response time: 10,703 ms]
▶ Data (56 bytes)

```

```

▶ Frame 51: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
▶ Ethernet II, Src: cc:01:0d:28:00:21 (cc:01:0d:28:00:21), Dst: Private_66:68:01 (00:50:79:66:68:01)
▶ Internet Protocol Version 4, Src: 192.168.1.2, Dst: 192.168.2.3
▼ Internet Control Message Protocol
  Type: Echo (ping) request (8)
  Code: 0
  Checksum: 0xbb18 [correct]
  [Checksum Status: Good]
  Identifier (BE): 25841 (0x64f1)
  Identifier (LE): 61796 (0xf164)
  Sequence Number (BE): 2 (0x0002)
  Sequence Number (LE): 512 (0x0200)
  [Response frame: 52]
▶ Data (56 bytes)

```

Когда PC1 из сети 100.170.10.0/24 пытается пинговать ПК2 из сети 192.168.10.0/24, он понимает, что получатель находится в другой подсети, поэтому пакет надо отправлять на шлюз по умолчанию. Для этого ПК1 сначала выполняет ARP-запрос для IP-адреса шлюза (100.170.10.200), чтобы узнать MAC-адрес интерфейса маршрутизатора. После получения ответа ПК1 отправляет ICMP-запрос на MAC-адрес маршрутизатора. Маршрутизатор, получив пакет, смотрит в свою таблицу маршрутизации и определяет, что для сети 192.168.10.0/24 пакет нужно отправить через другой интерфейс. На этом интерфейсе также выполняется ARP-запрос для IP-адреса ПК2, затем ICMP-запрос доставляется ПК2, а ответ идёт в обратном порядке. Важно отметить, что при отправке ICMP-ответа ПК2 также сталкивается с тем, что целевой IP (ПК1) находится в другой сети, поэтому он не может отправить кадр напрямую. Вместо этого ПК2 выполняет ARP-

запрос для своего шлюза по умолчанию (192.168.10.200), чтобы узнать его MAC-адрес. Именно этот ARP-запрос с Target IP 192.168.10.200 перехватывается на втором линке (между ПК2 и роутером) и виден на скриншоте. Таким образом, на каждом линке ARP-кадры всегда направлены на поиск локального шлюза, а не конечного получателя. На каждом линке можно перехватить соответствующие ARP- кадры и ICMP-пакеты, что позволяет проследить полный путь данных.